

SCI-Gaussian: Optimizing 3D Gaussian Radiance Fields from a Snapshot Compressive Image

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Abstract—Snapshot compressive imaging (SCI) is a compressed sensing (CS)-based high-speed imaging modality. Recent efforts have explored the underlying 3D representation from only an SCI image using neural radiance fields (NeRF), yet the training time, rendering computation cost, and reconstruction quality limitations are general issues that have limited wider adoption. This paper introduces SCI-Gaussian, the first 3D-aware SCI reconstruction based on 3D Gaussian splatting (3D-GS). This method utilizes an explicit 3D representation to achieve efficient and high-quality scene reconstruction. The motivation stems from the highly efficient representation and surprising quality of 3D-GS, despite when applied to SCI system, it encounters difficulties in generating point initialization for explicit Gaussians and accurate pose recovery from a single SCI measured image. Specifically, we effectively initialize these Gaussians through sampling a coarsely trained NeRF at various hash structures, then model the physical formation of the SCI measurement and jointly optimize Gaussians and camera trajectories with a bundle adjustment formulation during exposure time. Extensive experiments on synthetic and real-world datasets demonstrate that SCI-Gaussian outperforms the state-of-the-art (SOTA) methods, achieving comparable or better results with significantly 10× faster training and 1000× faster rendering speed than the most recent NeRF-based method.

Index Terms—3D Gaussian Splatting, Snapshot Compressive Image, Differentiable Rendering.

I. INTRODUCTION

Inspired from compressed sensing (CS) [1], [2], snapshot compressive imaging (SCI) [3] has emerged to address issues such as hardware cost and storage demands associated with high-speed imaging systems. An SCI system usually contains a hardware encoder employing a series of specially designed 2D masks to compress the incoming photons across exposure time into a single image, and the software decoder restores the higher framerate images as shown in Fig. 1. While several

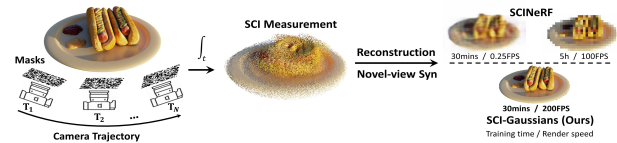


Fig. 1. **SCI-Gaussian vs. SCINeRF.** Our method captures extremely high-frequency details and achieves a significant training and rendering speedup compared to the SOTA.

algorithms have been proposed, including model-based methods [4]–[6] as well as deep-learning methods [7]–[11], for SCI reconstruction, they can only recover images corresponding to the applied masks and limited work has been proposed to consider the underlying 3D structure to ensure multi-view consistency. Recently, neural radiance fields (NeRF) [12] have been employed to SCI and SCINeRF [13] is one attempt to recover the 3D representation within the exposure time. However, the method encounters significant challenges in achieving low training cost, real-time rendering, and producing high-quality outputs with fine detail.

A method under active study, 3D Gaussian splatting (3D-GS) [14], [15] extends implicit neural rendering to explicit 3D Gaussians. By projecting Gaussians onto the image plane, 3D-GS achieves real-time rendering with improved rendering quality. These properties position 3D-GS as a compelling alternative to NeRF-based methods. **However, the challenges posed by utilizing 3D-GS to SCI can be attributed to three primary factors:** (a) 3D-GS relies on multi-view sharp images for supervision, while SCI returns only one single compressed image; (b) accurate camera poses are essential for training 3D-GS, but it is impossible to recover poses from only one SCI image using COLMAP [16]; (c) vanilla 3D-GS requires a point cloud from COLMAP as the initialization.

In order to address these challenges, we propose Snapshot Compressive Image Gaussian Splatting, the first SCI framework based on 3D-GS, dubbed SCI-Gaussian. To initiate the

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point cloud of the Gaussians, we replace vanilla NeRF from SCINeRF to a coarse grid-based NeRF [17] which enables the rapid creation of an initial and rough 3D modeled scene. In contrast to COLMAP, which exhibits very poor performance with few captured images and cannot work when fewer than two, our approach is relatively unaffected by the number of captured images. We subsequently sample the grid-based NeRF to form a corresponding point cloud for the initialization of SCI-Gaussian. We incorporate the physical imaging model of the SCI system and the camera trajectory function into the training of 3D-GS, and jointly optimize camera trajectory and Gaussian themselves, i.e., bundle-adjusted formulation [13], within the exposure time.

SCI-Gaussian is evaluated on both synthetic and real datasets. The results demonstrate that it outperforms prior SOTAs. In summary: **(a)** We propose the first real-time 3D-aware reconstruction and novel-view synthesis from a single SCI image; **(b)** we introduce a bundle-adjusted SCI framework specifically which can jointly optimize the camera trajectory as well as the 3D-GS parameters; **(c)** our approach successfully synthesizes higher-quality 3D-aware novel-view images and achieves $10\times$ faster training and $1000\times$ faster rendering, surpassing the most recent SOTAs.

II. METHODOLOGY

As shown in Fig. 2, SCI-Gaussian aims to recover a sharp 3D scene representation by jointly learning the camera motion trajectories and the parameters of Gaussian, given a single SCI measured image and corresponding masks without any detailed pose information.

A. Preliminaries

1) *Video SCI Image Formation Model*: The process of an SCI system is similar to that of the image restoration from a blurry image. The difference is that each input image is modulated by a unique binary mask across the exposure time. The whole imaging process can be described formally as:

$$\mathbf{Y} = \sum_{i=1}^N \mathbf{X}_i \odot \mathbf{M}_i + \mathbf{Z}, \quad (1)$$

where $\mathbf{Y}, \mathbf{X}_i \in \mathbb{R}^{H \times W}$, $\mathbf{M}_i \in \mathbb{R}^{H \times W}$ are the captured SCI image, the i th virtual image and corresponding binary mask within exposure time respectively, N is the compression ratio (equivalent to the number of virtual images), H and W are image height and width respectively, $\odot$ denotes element-wise multiplication, and $\mathbf{Z} \in \mathbb{R}^{H \times W}$ is the measurement noise. The individual pixel value in the binary mask is randomly generated. Given the explicit 3D-GS representation and corresponding camera poses, we are able to render a series of virtual image $\mathbf{X}_i$ to synthesize the compressed image $\mathbf{Y}$. We can see that $\mathbf{Y}$ is differentiable with respect to Gaussian and the poses, which lays the foundation for the optimization.

B. Proposed Framework

1) *Point Cloud Initialization from NeRF*: SCI for one scene only has one captured measurement and it is impossible to recover the camera poses and generate the initialized point

via COLMAP. It has also been observed that 3D-GS struggles to reconstruct from random points or points that are distant from the scene's center [18], which critically impairs 3D-GS's capacity to model objects. In contrast, NeRF-based methods [13] offer a more efficient and robust approach for point cloud generation that is largely independent of the size of the training set. Motivated by this insight, instead of obtaining the initialization from COLMAP, we alternative to first run the modified SCINeRF.

As vanilla NeRF frameworks always encounter the challenges with long training and render time while our framework aims to mitigate these limitations. Specifically, for the initialization, we modify SCINeRF by substituting MLP with hash maps [17] to make it converge faster. Additionally, since we just need to model the scene coarse structure, this accelerated training process only takes approximately 2000 iterations and around 10 minutes. Then, to generate points from the trained NeRF model F_Φ , we cast rays from the trained cameras and sample points at the estimated termination point $\hat{z}$ of each ray. The position and color of each generated point are obtained as:

$$x_i = \mathbf{o} + \hat{z}\mathbf{d}, \quad c_i = F_\Phi(x_i, \mathbf{d}), \quad (2)$$

where $\mathbf{o}$ and $\mathbf{d}$ represent the origin and direction of the ray, respectively. We sample N_P points from the coarse NeRF to form the point cloud, denoted by $P = \{(x_i, c_i) \mid 1 \leq i \leq N_P\}$.

2) *Camera Motion Trajectory Modeling in 3D-GS*: 3D-GS represents scenes as a collection of Gaussians $\mathbf{G}$ initialized from N_P points using the same way in original 3D-GS [14]. We denote the position, opacity, and color of the i th Gaussian by x_i , α_i , and c_i . To render an image from a given viewpoint $\mathbf{T}_c$, Gaussians are projected onto the image plane and then sorted in accordance with their depth values. The rendered color $\hat{\mathbf{C}}$ for a pixel is determined by blending all Gaussians that are projected onto it in order, according to

$$\hat{\mathbf{C}} = \sum_{i=1}^N c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (3)$$

demonstrating that the rendered pixel color is differentiable with respect to all of the learnable Gaussians, i.e., $\mathbf{G}_\theta$, and the camera poses $\mathbf{T}_c$. This facilitates the bundle adjustment formulation, accommodating an SCI image related to a series of high-speed frames and initialized inaccurate camera poses within the framework of 3D-GS. In detail, by assuming the exposure time is typically short, we can linearly interpolate the camera pose of initial frame $\mathbf{T}_{\text{start}}$ and final frame $\mathbf{T}_{\text{end}}$ to obtain every virtual pose along the trajectory. This estimation is accomplished by minimizing the following loss function, which includes an $\mathcal{L}_1$ loss and a D-SSIM loss between $\hat{\mathbf{Y}}$, the synthesized SCI image from 3D-GS using the measurement formation model described by Eq. (1), i.e. $\hat{\mathbf{Y}} = \sum_{i=1}^N \mathbf{C}_i \odot \mathbf{M}_i$, and $\mathbf{Y}$, the captured SCI measurement:

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_1 + \lambda\mathcal{L}_{\text{D-SSIM}}. \quad (4)$$

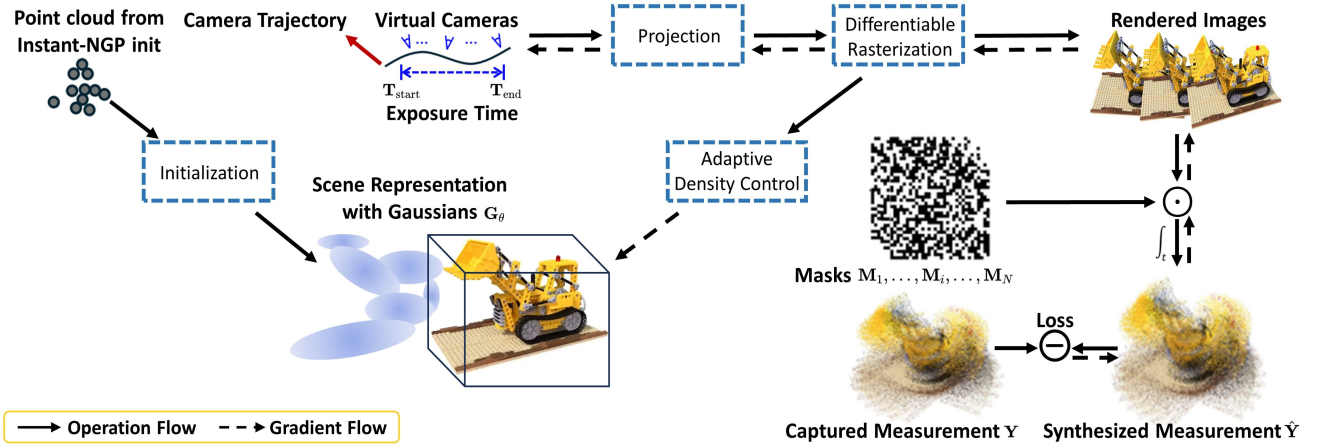


Fig. 2. **Overview of the proposed SCI-Gaussian.** Our method takes a single snapshot compressed image and corresponding masks as input, and recovers the underlying 3D scene representation as well as the camera motion trajectory within an exposure time in real time.

To optimize the learnable Gaussian parameters θ , and camera pose $\mathbf{T}$ for each virtual image, it is necessary to derive the Jacobians and complete the gradient flow:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} \cdot \frac{1}{n} \sum_{i=1}^N \frac{\partial \hat{\mathbf{Y}}}{\partial \hat{\mathbf{C}}_i} \frac{\partial \hat{\mathbf{C}}_i}{\partial \theta}, \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{T}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} \cdot \frac{1}{n} \sum_{i=1}^N \frac{\partial \hat{\mathbf{Y}}}{\partial \hat{\mathbf{C}}_i} \frac{\partial \hat{\mathbf{C}}_i}{\partial \theta} \frac{\partial \theta}{\partial \mathbf{T}}, \quad (6)$$

where $\hat{\mathbf{C}}_i$ denotes i th rendered image from 3D-GS, and we parameterize $\mathbf{T}_{\text{start}}$ and $\mathbf{T}_{\text{end}}$ as 4×4 matrices in $\text{SE}(3)$.

III. EXPERIMENTS

A. Experimental settings

Datasets. We generate synthetic datasets from scenes provided in DeblurNeRF [19] with Blender. There are four virtual scenes in the datasets, including *Cozy2room*, *Vender*, *Tanabata*, and *Factory*, and the resolution is 600×400 . For real-world dataset, we collect SCI measurement from a real video SCI setup. The setup consists of an iRAYPLE A5402MU90 camera and a FLDIS COVERY F4110 DMD. The resolution is 1024×768 .

Baselines and evaluation metrics. We compare SCI-Gaussian with SOTA SCI methods, including GAP-TV [4], PnP-FFDNet [5], EfficientSCI [11] and SCINeRF [13]. Widely used metrics including structural similarity index (SSIM), peak signal-to-noise ratio (PSNR), and learned perceptual image patch similarity (LPIPS) are employed for quantitative comparisons.

Implement details. We implement with PyTorch on a NVIDIA RTX 3090 GPU. The framework is first initialized with instant-NGP [17]. Both weight network and color network are with an MLP, containing 2 hidden layers with a width of 64 neurons. We use the Adam optimizer with a learning rate from 1×10^{-4} to 1×10^{-5} and use the same hyper-parameter setting for hash encoder as [17], with batchsize of 5000 for

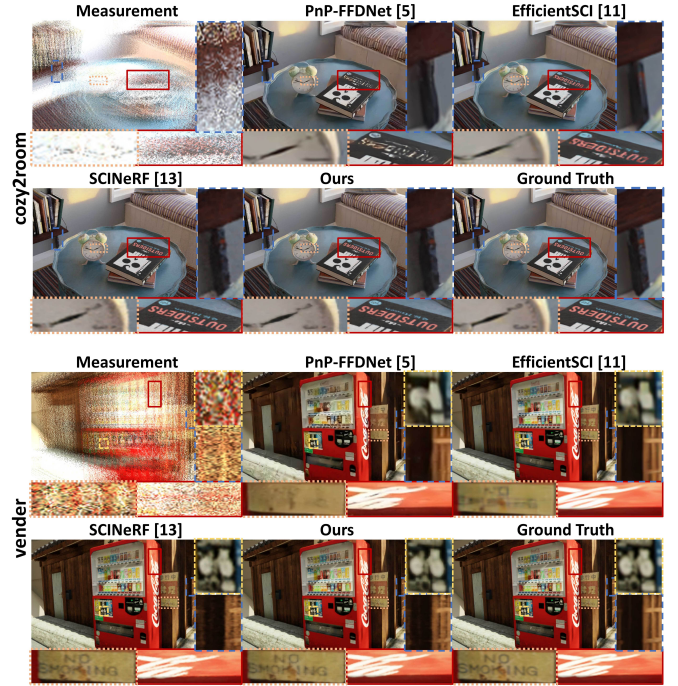


Fig. 3. **Qualitative evaluations of our method against SOTA SCI image restoration methods on synthetic dataset.** Due to space limitations, two samples from *Cozy2room* and *Vender* are selected for visualization.

one iteration. The learning rate for optimizing Gaussians is identical to original 3D-GS [14] and for the estimated poses decreases exponentially from 1×10^{-3} to 1×10^{-5} . The compression ratio of benchmark is 8. We train our method 2.5k iterations with around 10 minutes for initialization and 20k iteration with around 20 minutes training time for optimizing trajectory and Gaussians for 3D scenes.

B. Results

Performance on synthetic dataset. The qualitative results and visual examples for SCI reconstruction on series of synthetic dataset are reported in Table I and Fig. 3. From both

TABLE I
QUANTITATIVE SCI RECONSTRUCTIONS ON THE SYNTHETIC DATASETS. **BOLDFACE**: BEST, UNDERLINE: SECOND BEST.

Method	Cozy2room			Vender			Tanabata			Factory		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
SCI-Gaussian (Ours)	34.68	0.9674	<u>0.0472</u>	37.89	0.9874	0.0235	36.82	0.9872	0.0308	<u>35.98</u>	0.9743	<u>0.0257</u>
GAP-TV [4]	21.77	0.4321	0.6031	20.00	0.3678	0.6882	20.42	0.4264	0.6250	24.05	0.5666	0.5149
PnP-FFDNet [5]	28.98	0.8916	0.0984	28.70	0.9235	0.1315	29.17	0.9032	0.1197	31.75	0.8977	0.1142
EfficientSCI [11]	31.47	0.9327	0.0476	33.17	0.9401	0.0456	32.30	0.9587	0.0600	32.87	0.9259	0.0709
SCI-NeRF [13]	<u>33.23</u>	<u>0.9492</u>	0.0445	<u>36.40</u>	<u>0.9840</u>	<u>0.0298</u>	<u>33.61</u>	<u>0.9638</u>	<u>0.0374</u>	36.60	<u>0.9638</u>	0.0221

TABLE II
QUANTITATIVE COMPARISONS ON NOVEL-VIEW SYNTHESIS USING THE SYNTHETIC DATASETS.

Method	Cozy2room			Vender			Tanabata			Factory		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
SCI-Gaussian (Ours)	34.19	0.9598	<u>0.0481</u>	36.37	0.9792	0.0225	36.28	0.9789	<u>0.0301</u>	<u>35.25</u>	0.9627	0.0342
NeRF+GAP-TV [4]	21.99	0.5027	0.5212	21.68	0.4365	0.5536	20.91	0.4122	0.5531	25.48	0.6853	0.4401
NeRF+PnP-FFDNet [5]	30.15	0.9030	0.0919	30.02	0.9343	0.1317	28.00	0.9025	0.1400	31.96	0.8850	0.1823
NeRF+EfficientSCI [11]	31.79	0.9357	0.0641	32.77	0.9431	0.0648	31.35	0.9535	0.0769	32.72	0.8942	0.0676
SCINeRF [13]	<u>33.12</u>	<u>0.9477</u>	0.0357	<u>34.75</u>	<u>0.9744</u>	<u>0.0257</u>	<u>32.84</u>	<u>0.9574</u>	0.0297	<u>36.33</u>	<u>0.9602</u>	<u>0.0426</u>

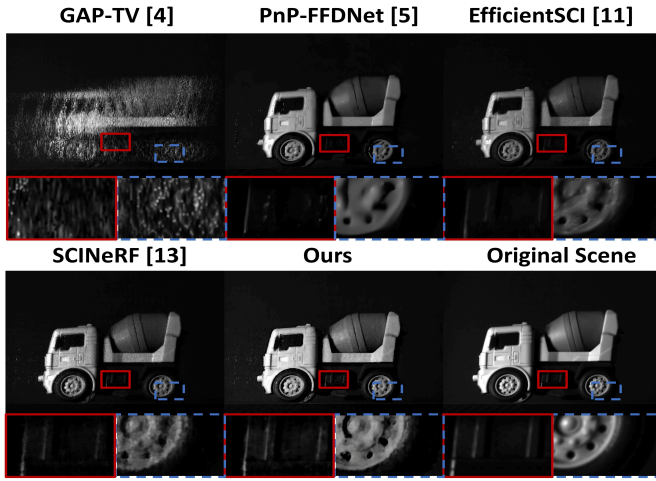


Fig. 4. **Qualitative evaluations of SCI-Gaussian against SOTA SCI image restoration methods on the real-world dataset.** Since the ground truth in real datasets are unavailable, we capture separate scenes after obtaining the SCI reconstruction used for reference.

the quantitative and qualitative results, we can see that SCI-Gaussian restores finer details and more precise geometries such as shown in the zoomed-in portions related to textures or characters. Additionally, we compare the performance on novel-view synthesis task against the recent SCINeRF as well as the vanilla NeRF using the reconstructed images from other comparison methods and the estimated poses using ground truth as input, biasing the experiment against our method. However, SCI-Gaussian still achieves excellent performance and corresponding quantitative results are given in Table II.

Performance on real-world dataset. For qualitative comparisons on real datasets, SCI-Gaussian renders the images from the learned 3D representation and demonstrates that it can reconstruct the encoded images with correct geometry and significantly fewer defects compared to SOTAs, as shown in Fig. 4. Of particular interest is the observation that SCINeRF seems to encounter difficulties with exposure offsets, whereas SCI-Gaussian consistently achieves effective correction.

TABLE III
COMPARISON ON EFFICIENCY OF DIFFERENT METHODS, REPRESENTED IN TRAINING TIME (HOUR) AND RENDERING SPEED (FPS).

Method	EfficientSCI [11]		SCINeRF [13]		SCI-Gaussian (Ours)	
	Time	FPS	Time	FPS	Time	FPS
	> 120h	2.6	~ 5h	0.25	~ 0.5h	200

TABLE IV
ABLATION STUDIES WITH DIFFERENT COMPRESSION RATIOS, REPRESENTED AS THE MEAN METRICS UNDER SYNTHETIC DATASETS.

	CR16		CR24		CR32	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
SCI-Gaussian (Ours)	35.27	0.9614	34.48	<u>0.9375</u>	<u>31.34</u>	<u>0.8984</u>
GAP-TV [4]	19.50	0.4975	18.63	0.4700	18.25	0.4425
PnP-FFDNet [5]	24.35	0.7950	22.21	0.7325	23.55	0.6975
EfficientSCI [11]	31.60	0.9093	29.10	0.8717	28.43	0.8443
SCI-NeRF [13]	<u>34.88</u>	<u>0.9600</u>	<u>34.37</u>	0.9532	33.95	0.9481

Efficiency. We further compare the efficiency by calculating the average training time and rendering speed on the synthetic datasets. Table III shows that our method has 10× faster training and 1000× faster rendering speed with comparable or better performance compared to SOTAs, indicating the potential of SCI-Gaussian in practical applications.

Ablation study for different compression ratios. We also study the performance of SCI-Gaussian under different compression ratios. The results demonstrate the stable performance of SCI-Gaussian under high compression ratios, even though it is slightly outperformed by NeRF-based method.

IV. CONCLUSION

We propose the first 3D-aware SCI reconstruction based on 3D-GS. We validate the proposed method on synthetic and real-world datasets, verifying its high-quality performance for multi-view consistency reconstruction, extremely low training cost and real-time rendering compared to the most recent SOTAs, which significantly provides a possibility for practical deployment of 3D-aware SCI such as for real-time private transmission between edge devices and cloud infrastructure.

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